19. Two mirrors are at an angle θ. A ray incidenting at a certain angle is undergoing two reflections, then the angle of deviation is

$\alpha = 2\theta$ if θ is acute

$\alpha = (2\pi - \theta)$ if θ is obtuse

Note : The angle of deviation is same for any angle of incidence as long as θ remains same.

20. If the mirror is shifted through a distance 'x' towards a stationary objectits image shifts through a distance '2x' towards the object.

21. If the mirror shifts through a distance 'x' towards an object and the object also shifts through 'y' towards the mirror then the image shifts through (2x+y) towards the mirror.

22. When a person moves towards a mirror with a velocity "V" then
a) His image approaches the mirror with the same velocity V.
b)He observes his image as if it is approaching him with a velocity 2V (ie.relative velocity)

23. When a mirror moves towards a person at rest with a veloc ity 'V'
a) His image moves with a velocity 2V.
b) His image approaches the mirror with a velocity V.
c) He observes his image to be approaching him with a velocity "2V" [ie. relative velocity]

24. When a mirror and a person are moving towards each other with same velocity V, then
a) His image moves with a velocity 3V.
b) His image appraoches the mirror with a velocity 3V-V = 2V.
c) He observes his image as if it is approaching him with a velocity 3V + V = 4V.

25. To see a complete wall behind himself , minimum height of mirror is one third of height of wall whenn observer in the middle of the wall and mirror

Image charachteristics

Image : The image of an object can be defined as the impression of the object created by the light on the mirror.

1. **Object Distance and Image Distance**

One of the characteristic of plane mirror images pertains to the relationship between the object's distance to the mirror and the image's distance to the mirror. For plane mirrors, the object distance is equal to the image distance That is the image is the same distance behind the mirror as the object is in front of the mirror.

ASSIGNMENT

1. Mass of electron is

   1) 9.1083×10^{-31} kg 2) 9.1083×10^{-24}kg

   3) 9.1083×10^{-28} kg 4) 1.67×10^{-24} gm

2. Rutherford's scattering experiment is related to

   1) neutrons 2) atom 3) election 4)nucleus

3. The electron is

   1) α –ray particle 2) β–ray particle 3) positron 4) γ–ray

4. The e/m ratio of an electron for all types of substances is

   1) constant 2) less than one 3) greater than one 4) zero

5. A proton is identical to

   1) the nucleus of helium 2) the nucleus of a hydrogen atom

   3) a molecule of a hydrogen 4) an atom of hydrogen

6. Isotopes have the same

   1) Atomic number 2) Electronic configuration

   3) Mass number-neutron 4) All of the above

7. Atom and ion differ in

   1) the relative size 2) the configuration

   3) the presence of charge 4) all of these

8. Value of e/m of an electron was measured by

   1) Millikan 2) J.J. Thomson 3)Rutherford 4) Dalton

9. The atomic number of an element is 11 and its mass number is 23. The perspective number of electrons, protons and neutrons in this atom will be

   1)11, 11, 12 2)11, 12, 11 3) 12, 11, 11 4) 23, 11, 23

10. The space between proton and electron in a hydrogen atom is

    1) absolutely empty 2) full of electromagnetic radiation

    3) full of air 4) full of ether

ATOMIC NUMEBRS OF FIRST THIRTY ELEMENTS

Atomic Number	Name of the element	Symbol
1	Hydrogen	H
2	Helium	He
3	Lithium	Li
4	Berylium	Be
5	Boran	B
6	Carbon	C
7	Nitrogen	N
8	Oxygen	O
9	Flourine	F
10	Neon	Ne
11	Sodium	Na
12	Magnesium	Mg
13	Aluminium	Al
14	Silicon	Si
15	Phosphorous	P

Atomic Number	Name of the element	Symbol
16	Sulphur	S
17	Chlorine	Cl
18	Argon	Ar
19	Potassium	K
20	Calcium	Ca
21	Scandium	Sc
22	Titanium	Ti
23	Vanedium	V
24	Chromium	Cr
25	Manganese	Mn
26	Iron	Fe
27	Cobalt	Co
28	Nickle	Ni
29	Copper	Cu
30	Zinc	Zn

Isotopes and Isobars

- Atoms of an element which differ in their mass numbers but have the same atomic number are called isotopes.
- Isotopes have same number of protons but differ in the number of neutrons present in them.

ASSIGNMENT KEY

01. 1	02. 4	03. 2	04. 1	05. 2	06. 4
07. 4	08. 2	09. 1	10. 1	11. 2	12. 3
13. 2	14. 1	15. 3	16. 3	17. 3	18. 3
19. 1	20. 3	21. 3	22. 2	23. 2	24. 2
25. 2	26. 1	27. 3	28. 3	29. 2	30. 1

SELF TEST

1. The nucleus of an atom contains
   1) Electrons and protons 2) Protons and neutrons
   3) Electrons and beta particles 4) Protons and alpha particles
2. The number of neutrons present in the radioactive isotope of hydrogen is
   1) 1 2) 2 3) 3 4) 0
3. Rutherford's scattering experiment led to the conclusion that?
   1) nucleus is very small in size compartive to the size of an atom
   2) the nucleus is very heavy and carries negative charge
   3) most of the space in the atom is empty
   4) all of these
4. The example of a pair of isobar is
   1) ${}_1H^1$ and ${}_1H^2$ 2) ${}_1H^2$ and ${}_1H^3$ 3) ${}_1H^3$ and ${}_2He^3$
   4) ${}_1He^3$ and ${}_1He^4$
5. Nuclei of isotopes differ in
   1) number of protons 2) number of neutrons
   3) number of neutrons and protons both
   4) none of these
6. The number of unpaired electrons in a chromic conclusion Cr^{+3} (atomic number 24) is
   1) 6 2) 4 3) 3 4) 1

09. In your study room a plane mirror is fixed to the wall in front of you and opposite to that wall, a wall-clock is hanging in which only marking were there without numbers. You observe the time in the mirror as 7:10 hrs. Then what is the actual time.

1) 4:50 hrs 2) 7:10 hrs 3) 7:50 hr 4) 5:50 hrs.

10. A ray of light is incident on a plane mirror at an angle of 30^0 What is the angle of deviation?

1) 60° 2) 30° 3) 90° 4) 120°

11. On rotating a mirror through an angle of 20^0, the reflected ray is found to get deviated through an angle

1) 20^0 2) 10^0 3) 40^0 4) 60^0

12. A man of 180 cm height stands in front of a plane mirror. His eyes are at a height of 170 cm from floor. Then the minimum length of a plane mirror required for him to see his full image is

1) 90 cm 2) 180 cm 3) 45 cm 4) 360 cm

13. Two plane mirrors are inclined at 70^0. A ray incident on one mirror at an angle θ after reflection falls on the second mirror and is reflected from there parallel to the first mirror, then θ is

1) 50^0 2) 45^0 3) 30^0 4) 55^0

14. A ray deviates at 90^0 after suffering reflection from a mirror. The angle of incidence is

1) 90^0 2) 30^0 3) 60^0 4) 45^0

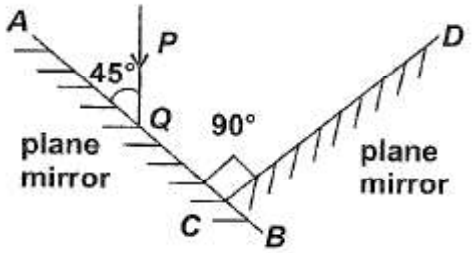
15. A man 1.8 m tall wishes to see full-length image in a plane mirrior. The length of the shortest mirror in which he can see his entire image is

1) 0.6 m 2) 0.9 m 3) 1.2 m 4) 1.5 m

16. A plane mirror is approaching you at 10 cm/s. You can see your image in it. The image will approach you with a speed of

1) 5 cm/s 2) 10 cm/s 3) 15 cm/s 4) 20 cm/s

17. Two plane mirrors AB and CD are placed at right angles to one another. A ray of light PQ hits AB at Q making an angle of 45° as shown in the diagram. This ray will be reflected from mirror CD at an angle.



1) 90^0 2) 30^0 3) 60^0 4) 45^0

18. A light ray is made to incident on a glass plate with angle of incidence 15°and then reflected. Then the angle of deviation is

1) 45° 2)130° 3) 150° 4) 90°

LIGHT

Introduction & Properties of Light

1. **Definition of Light:** It is an invisible energy which causes in us sensation of sight (vision). Since light is obtained from heat energy, i.e., when an object is heated to a temperature beyond (500 ^{0}C, we can say that light is a kind of energy). It must be kept in mind that light energy makes the surrounding objects visible, but by itself it is an invisible energy.

2. **Terms related to Light**

i) **Optical Medium:** Any material (or) non-material through which light energy passes wholly (or) partially is called optical medium

Examples: Vacuum, air, water, glass etc.

ii) **Homogeneous Medium:** An optical medium which has a uniform composition throughout is called homogeneous medium.

Examples*:* Vacuum, diamond, distilled water, pure alcohol, etc.

iii) **Heterogeneous Medium:** An optical medium, which has different composition at different points is called heterogeneous medium.

Examples*:* Air , muddy water, fog, mist, clouds, smoke etc.

iv) **Transparent Medium:** A medium which allows most of the light energy to pass through it is called transparent medium.

Examples: Vacuum, glass, clear air, alcohol, benzene, etc.

v) **Translucent Medium:** A medium which partially allows the light energy to pass through it is called translucent medium.

Examples*:* Oiled paper, tissue paper, ground glass, butter paper, etc.

vi) **Opaque Bodies:** The bodies which do not allow the light energy to pass through them are called opaque bodies.

Examples*:* Bricks, wood, metals, etc.

vii) **Point Source of light:** A source of light which is of the size of pinhead is called point source of light.

Examples*:* The pinhole act as a point source of light.

viii) **Extended Source of Light :** Any source of light which is bigger than point source of light is called extended source of light

Examples*:* Bulb, Tube light, burning candle etc.

ix) **Ray of Light:** The path along which light energy travels in a given direction is called ray of light.

x) **Beam of Light:** A collection of number of rays of light is called beam of light.

xi) **Parallel Beam:** When the rays of light travels parallel to each other, then the collection of such rays is called parallel beam.

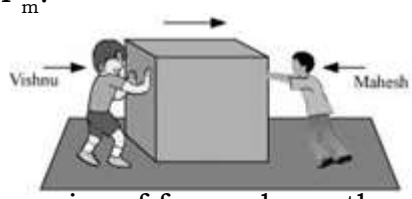
11. Positive electrification of a body is due to the
    1) Deficiency of protons 2) Excess of electrons
    3) Excess of nutrons 4) Deficiency of electrons
12. Which of the following device is used to find whether a given body is electrically charged or not ?
    1) gold leaf electroscope 2) microscope
    3) telescope 4) periscope
13. Lightining is caused by the
    1)Electrical discharge 2) heavy rain
    3) tides 4) none of these
14. Choose the correct statement
    1) gold leaf electroscope can be used to find whether a body has an electric charge on it or not.
    2) gold leaf electroscope can be used to find whether a body is positively charged or negatively charged.
    3) Both (1) and (2) 4) neither (1) nor (2)
15. The path through which an electric current flows is called
    1)Electric path 2) Electric circuit 3)Electric relay 4)Short circuit
16. Nichrome is an alloy. Which of the following metals make up this alloy?
    1) Nickel and chromium 2) Nitrogen and chromium
    3)Nitrogen, chlorine and chromium
    4)Nickel, chromium and manganese
17. What is the common voltage produced by a dry cell?
    1)1.5 V 2) 30 V 3) 60 V 4)3 V
18. In a dry cell, which of the following acts as the positive terminal?
    1) Carbon rod 2)Manganese dioxide
    3) Manganese dioxide and powdered carbon
    4) A metal cap on the carbon rod
19. Which of the following is the electrolyte in a dry cell?
    1)Copper sulphate 2) Zinc sulphate
    3)Sulphuric acid 4)Ammonium chloride
20. The electrolyte in a dry cell is ammonium chloride.Which of the following energy conversions takes place in a cell?
    1) Electrical energy into chemical energy
    2)Chemical energy into electrical energy

distances travelled by the boxes. They are related as $d_3 > d_2 > d_4 > d_1$.The sides a_1, a_2, a_3, and a_4 of the respective boxes are related as

1) $a_1 > a_2 > a_3 > a_4$ 2) $a_1 > a_4 > a_2 > a_3$
3) $a_3 > a_2 > a_4 > a_1$ 4) $a_4 > a_3 > a_2 > a_1$

10. Four similar boxes are labelled I, II, III, and IV. The masses of I and III are equal and the masses of II and IV are equal. The mass of box I is smaller than that of box II. Boxes I and II are placed in a grass field and boxes III and IV are placed on a marble floor.
Which box requires the maximum force to be moved?
1) I 2) II 3) II 4)IV

11. Vishnu and Mahesh are pushing a heavy box placed on a floor, as shown in the given figure. The arrow over the box shows the direction of motion of the box. Vishnu is applying a force of magnitude F_v and Mahesh is applying a force of magnitude F_m.



Which of the following pairs of forces have the same direction?
1) F_v and frictional force 2) F_m and frictional force
3) F_v and gravitational force 4) F_m and gravitational force

12. Ted learns about force of friction. He tries out different ways of reducing as well as increasing the friction on the surface of his driveway.
In order to increase the friction on the surface of the driveway, Ted should
1) Cover it with small pebbles 2) Cover it with wax
3) Pour water on it 4) Pour oil on it

13. Tony is playing carom along with his friends. After completing 2 games, some talcum powder is rubbed on to the board.
Talcum powder is used on the board to
1) Reduce its friction 2)Decrease its mass
3) Improve its fragrance 4)Improve its conductivity

14. Two boxes, having the same weight, are kept between two vertical walls. Calvin lifts one of the boxes (against the pull of gravitational force) with the help of a rope. Before lifting the second box, he lubricates the vertical walls with oil. As a result, he has to apply less force to lift the second box. Which force is reduced as a result of the lubrication of the walls?

1) Gravitational 2)Compressive 3)Frictional 4)Magnetic

15. John pushes a table with wheels under it. At the same time, Abraham pushes a table without any wheels under it. Both the tables weigh the same. The two tables move at the same speed, but John applies lesser force than Abraham.

Which force is reduced by the wheels under John's table?
1)Magnetic 2) Frictional 3) Electrostatic 4) Gravitational

1) 35 kmph 2) 210 kmph 3) 42 kmph 4) 30 kmph

15. During a journey ‘ a ’ travels x kilometers with velocity ‘ V ’ and the next ‘ x ’ kilometers with double the velocity. The average velocity during this journey is

1) 3V/2 2) 3V 3) $2V^2/3$ 4) 4V/3

16. An athelete completes one round of circullar track of radius R in 40ssecond.What will be the displacement at the end of 2min .20sec.

1) 2R 2) πR 3) 2πR 4) 6πR + R

17. A clock has its miniute hand 4.5 Cm long .Average velocity of the tip of the minute hand between 6.00 a.m to 6.15 a.m.

1) $\frac{1}{100\sqrt{2}}$ cm/sec2) $\frac{2}{100\sqrt{2}}$ cm/sec 3) 46.6 cm/sec 4) $4.5\sqrt{2}Cm/\sec$

18. While driving home from school you travel at 80 km/h for 120 km then slow to 40 km/h. You get home in 2 hours . How far is your hometown from school and what is the average speed?

1) 140km 2) 150km 3)200km 4)160km

19. A rock is thrown straight upward off the edge of a balcony that is 5 m above the ground. The rock rises 10 m, then falls all the way down to the ground below the balcony. What is the rock’s displacement?

1) 0 2) 5cm upward 3) 5 cm downward 4) 25cm

20. An athlete runs exactly once around a circular track with a total length of 500 m. Find the runner’s displacement for the race.

1) 0 2) 500cm 3) 250 cm 4) 25cm

21. A motor vehicle travelled the first third of a distance ‘s’ at a speed of v_1 = 10 kmph, the second third at a speed of v_2 = 20 kmph and the last third at a speed of v_3 = 60 kmph. Determine the mean speed of the vehicle over the entire distance ‘s’

1) 15 kmph 2) 12 kmph 3) 10 kmph 4)18 kmph

22. Body travels first $\frac{1}{3}$ of the distance with a velocity ‘u’, the second $\frac{1}{3}^{rd}$ with a velocity v and remaining distance with a velocity ω. The average velocity is

1) $\frac{u+v+\omega}{3}$ 2) $\frac{uv+v\omega+\omega\mu}{3}$ 3) $\frac{3uv\omega}{uv+v\omega+\omega u}$ 4) $\frac{uv+v\omega+\omega u}{uv\omega}$

23. If a cyclist takes one minute to complete half revolution on a circular path 120 m radius. What is the average velocity

1) 1 m/s 2) 2 m/s 3) 3 m/s 4) 4 m/s

24. One car moving on a straight road covers one third of the distance with 20 kmph and rest with 60 kmph. The average speed is

When arranged in the descending order of their magnitudes, the speed of sound in the three different media is

1) $v_s > v_l > v_g$ 2) $v_s > v_g > v_l$ 3) $v_g > v_s > v_l$ 4) $v_l > v_g > v_s$

8. The sound coming from an alarm clock cannot be heard if it is kept inside a box that is

1) filled with liquid only 2) filled with both solid and liquid

3) made up of iron with air inside it

4) devoid of any material medium inside

9. What should a hearing-impaired person wear in order to enhance his hearing power by amplifying sound?

1) Ear-horn 2) Stethoscope

3) Mobile headset 4)Jackhammer's headphone

10. The given table lists the main parts of the human ear that take part in hearing.Parts of the human ear

I Ear canal

II Three tiny bones

III Eardrum

IV Cochlea

The order in which sound vibration moves through the human ear is

1) I $\rightarrow$ II $\rightarrow$ III $\rightarrow$ IV 2) I $\rightarrow$ III $\rightarrow$ II $\rightarrow$ IV

3) III $\rightarrow$ I $\rightarrow$ IV $\rightarrow$ II 4) III $\rightarrow$ IV $\rightarrow$ I $\rightarrow$ II

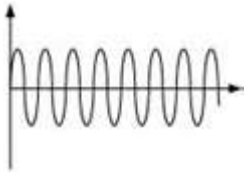
11. Vibration can be defined as the

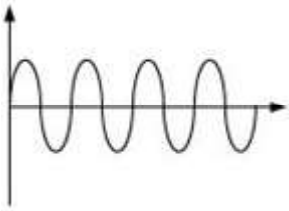
1) stretching of a membrane 2) roughness of a membrane

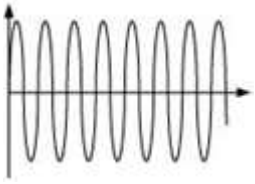
3) circular motion of a membrane

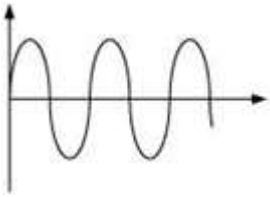
4) to and fro motion of a membrane

12. Which of the following sound waves produces the loudest sound?

1) 

2) 

3) 

4) 

13 With reference to sound, the number of to and fro motions of a membrane in per unit time is expressed in the unit of

1)angstrom 2)second 3)meter 4) hertz

Time	Second	S
Temperature	Kelvin	K
Luminous intensity	Candela	Cd
Electric charge	Coulomb	C
Current	Ampere	C
Amount of substance	Mole	Mole

Supplementary quantities

Angle	Radian	rd
Solid angle	Steridian	sr

17. **Multiples of Ten**

The study of nature involves objects of small size like atom, nucleus etc. They belong to the small world (microcosm).

If we measure the diameter of hydrogen atom (in microcosm) it will be described by a very small number 0.000,000,000,106m.

It also involves celestial objects of the universe like the sun, planets, moon, stars, etc.
They belong to the cosmic world (macrocosm).

The diameter of the sun will be described by a very large number 1,390,000,000m.

The above numbers are exceptionally big and difficult to write or remember. How then can this task be simplified?
It can be simplified by expressing in terms of multiples of 10.

The following chart shows some multiples of 10 and their corresponding names or prefixes :

Multiple and sub multiple factors

Multiplication factor	Name	Symbol
10^{12}	**tera**	**T**
10^{9}	**giga**	**G**
10^{6}	**mega**	**M**
10^{3}	**kilo**	**k**
10^{2}	**hecto**	**h**
10	**deca**	**da**
10^{-1}	**deci**	**d**
10^{-2}	**centi**	**c**
10^{-3}	**milli**	**m**

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FOUNDATION SCIENCE

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